PCA

$$X = \begin{array}{c} \begin{array}{ccc} x_1 & x_2 & x_3 \end{array} \\ \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 3 & 1 & 2 \\ 1 & 3 & 2 \end{bmatrix} \end{array} \qquad d = 3$$

reduce dimensionality → i) 3D → 2D.
ii) 3D → 1D

step 1 : mean centering.

$$\bar{u} = [1.5 \quad 1.5 \quad 1.5]$$

$$X_c = \begin{bmatrix} 0.5 & -1.5 & -0.5 \\ -1.5 & 0.5 & -0.5 \\ 1.5 & -0.5 & 0.5 \\ -0.5 & 1.5 & 0.5 \end{bmatrix}$$

step 2 : Compute covariance matrix.

$$\begin{bmatrix} cov(x_1, x_1) & cov(x_1, x_2) & cov(x_1, x_3) \\ cov(x_2, x_1) & cov(x_2, x_2) & cov(x_2, x_3) \\ cov(x_3, x_1) & cov(x_3, x_2) & cov(x_3, x_3) \end{bmatrix}$$

$$C = \frac{1}{n} X_c^T X_c$$

$$= \begin{bmatrix} 1.25 & -0.75 & 0.5 \\ -0.75 & 1.25 & 0.5 \\ 0.5 & 0.5 & 0.25 \end{bmatrix}$$

Steps: Compute eigenvalues & eigenvectors

$\lambda_1 \approx 2.0$, $\lambda_2 \approx 0.75$, $\lambda_3 = 0.0$

eigenvectors:-

$$u_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$u_2 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

$$v_3 = \frac{1}{\sqrt{3}} \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$$

Dimensionality reduction

3D → 2D

Take top 2 eigenvectors

$$W_{2D} = [v_1 \ v_2]$$

Project the data points

$$Z = X_c W_{2D} \qquad Z = [\ ]$$

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- each point will be of 2D

Take top 1 eigenvector

$W_{10} = [v_1]$

Total variance :-

$= 2 + 0.75 + 0$

$= 2.75$

$PCA = \frac{2}{2.75} = 73\% \Rightarrow PC1$

$PC2 = \frac{0.75}{2.75} = 27\%$

$PC3 = 0$